Tamarin in 90 Minutes

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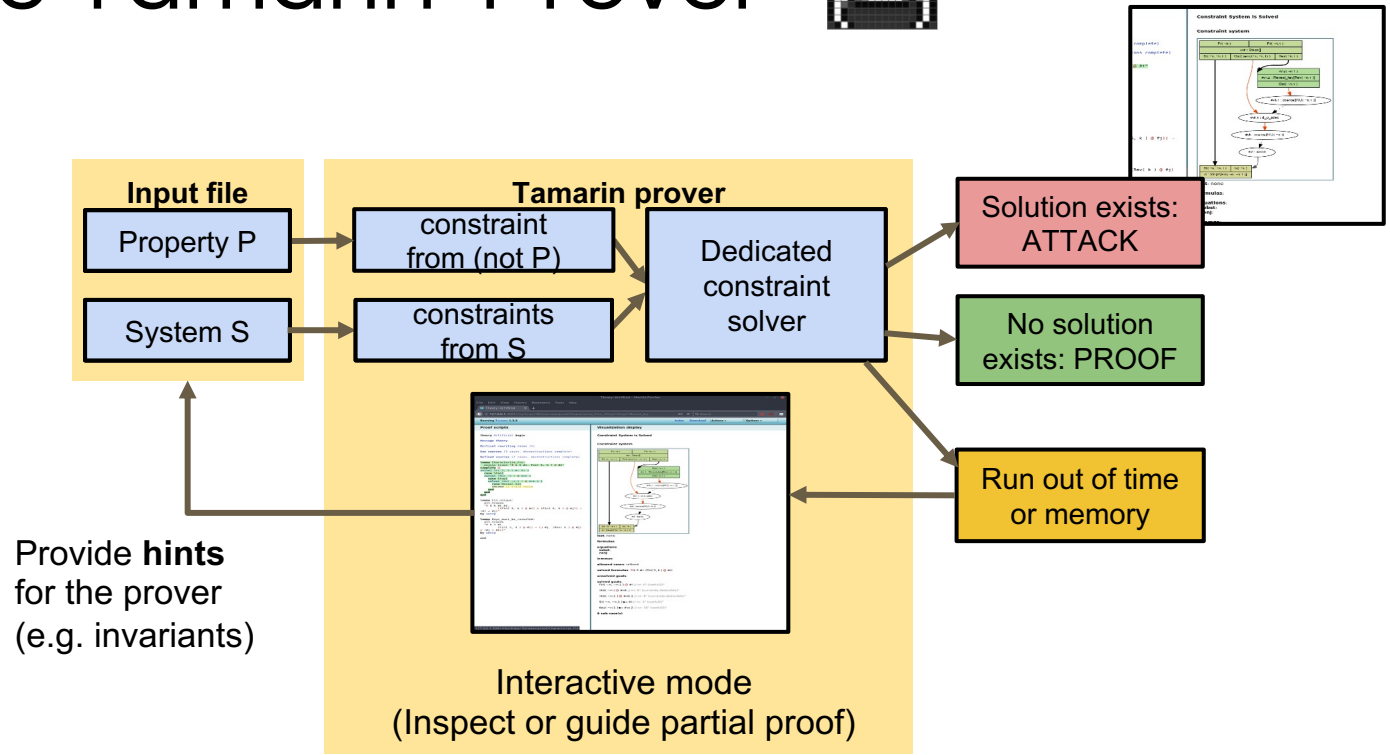
Slides designed by Cas Cremers, David Basin, Jannik Dreier, and Ralf Sasse

Sources:
Tamarin picture used with chicken hat by Brocken Inaglory
All other Tamarin photographs by Martin Dehnel-Wild
Other photos, graphics, and chicken hats by Cas Cremers

What to expect today?

- Introduction and small demo of the tool
- Some exercises to try it out yourself

The Tamarin Prover



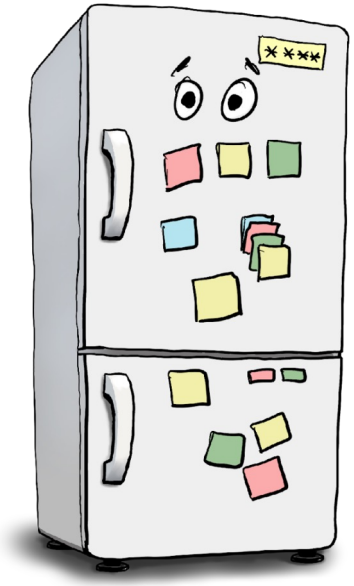
Demo

Modeling in Tamarin

Modeling in Tamarin

- Basic ingredients:
  - **Terms** (think “messages”)
  - **Facts** (think “sticky notes on the fridge”)
  - Special facts: **Fr(t)**, **In(t)**, **Out(t)**, **K(t)**
- State of system is a multiset of facts
  - **Initial state** is the empty multiset
  - **Rules** specify the transition rules (“moves”)
- Rules are of the form:

  l --> r or l --[a]-> r



The model

- **Term algebra**
  - enc(_,_), dec(_,_), h(_,_), _^_, _$^{-1}$, _*_, 1, …
- **Equational theory**
  - dec(enc(m,k),k) $=_E$ m,
  - (x^y)^z $=_E$ x^(y*z),
  - $(x^{-1})^{-1} =_E$ x, ...
- **Facts**
  - F(t1,...,tn)
- **Transition system**
  - State: multiset of facts
  - Rules: l –[a]→ r
- **Tamarin-specific**
  - Built-in Dolev-Yao attacker rules: In( ), Out( ), K( )
  - Special **Fresh** rule:
    - [] --[]--> [Fr(**x**)]
      - Constraint on system such that **x** is unique

Semantics

- **Transition relation**

  $S \text{–}[a]\rightarrow_R \ ((\ S \setminus^{\#} l\ ) \cup^{\#} r\ )$, where

  - $l \text{–}[a]\rightarrow r$ is a ground instance of a rule in R, and
  - $l \subseteq^{\#} S$ wrt the equational theory

- **Executions**

  $\mathrm{Exec}(\ R) = \{\ [\ ] \text{–}[a_1]\rightarrow \ldots \text{–}[a_n]\rightarrow S_n \mid \forall n\ .\ \mathrm{Fr}(n)$ appears only once on right-hand side of rule $\}$

- **Traces**

  $\mathrm{Traces}(\ R) = \{\ [a_1,\ldots,a_n] \mid [\ ] \text{–}[a_1]\rightarrow \ldots \text{–}[a_n]\rightarrow S_n \in \mathrm{Exec}(\ R)\ \}$

‘c’ constant

Semantics: example 1

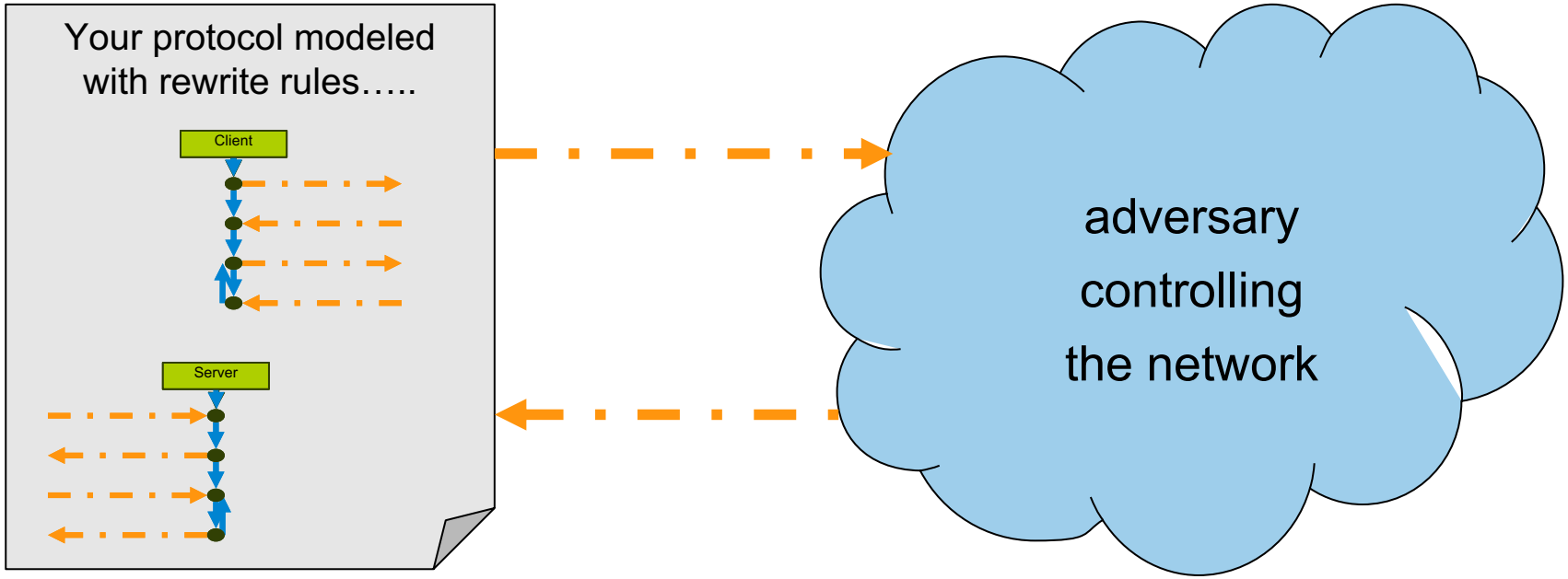
- **Rules**
  - rule 1: [ ] –[Init()]→ [A('5')]
  - rule 2: [A(x)] –[Step(x)]→ [B(x)]
- **Execution example**
  - [ ]
  - –[Init()]→ [A('5’)]
  - –[Init()]→ [A('5'), A('5')]
  - –[Step('5')]→ [A('5'), B('5')]
- **Corresponding trace:** [Init(), Init(), Step('5')]

‘c’ constant

Semantics: example 2 (persistent facts)

- **Rules**
  - rule 1: [ ] –[Init()]→ [!C('ok'), D('1')]
  - rule 2: [!C(x), D(y)] –[Step(x,y)]→ [D(h(y))]
- **Execution example**
  - [ ]
  - –[Init()]→ [!C('ok'), D('1')]
  - –[Step('ok','1')]→ [!C('ok'), D(h(’1’))]
  - –[Step('ok',h('1'))]→ [!C('ok'), D(h(h('1')))]
- **Corresponding trace:** [Init(), Step('ok', '1'), Step('ok', h('1'))]

Tamarin tackles complex interaction with adversary



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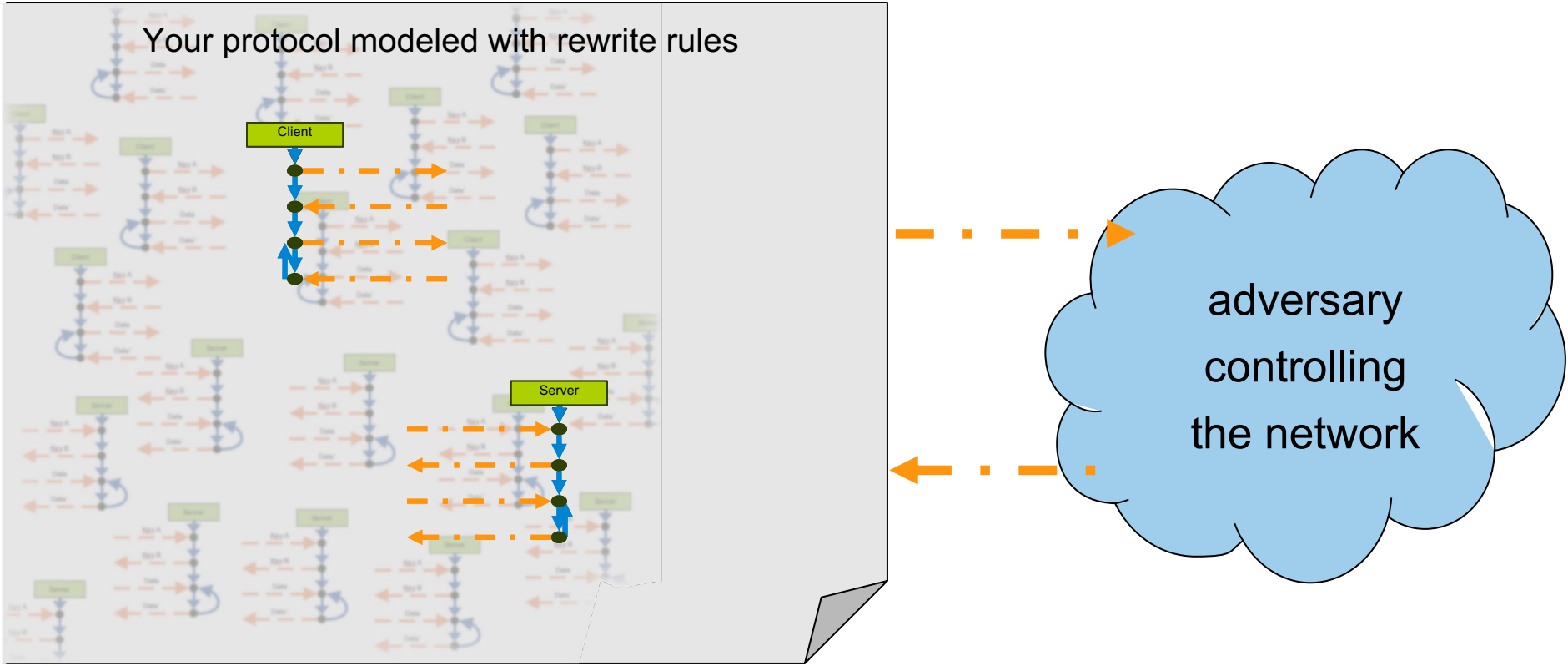

The NAXOS protocol

The Naxos protocol

lkA	A's long-term priv. key
g^lkA	A's long-term pub. key
eskA	A's eph. priv. key

$\boxed{\mathrm{I}}$		$\boxed{\mathrm{R}}$
Fresh esk_I $ex_I = h1(esk_I, lk_I)$ $hk_I = g^{ex_I}$	$\xrightarrow{hk_I}$	receive X Fresh esk_R $ex_R = h1(esk_R, lk_R)$ $hk_R = g^{ex_R}$
receive Y	$\xleftarrow{hk_R}$	

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$$key = h2(g^{(ex_R)(lk_I)}, g^{(ex_I)(lk_R)}, g^{(ex_I)(ex_R)}, I, R)$$

Modeling Naxos

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```
rule Init_1:
  let exI = h1(<~eskI, ~lkI >)
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   [ Fr( ~eskI ) ] --> [ Out( hkI ) ]
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rule generate_ltk:
   let pkA = 'g'^~lkA
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rule Init_2:
   [ In( Y ) ] --> []
```

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   [ Fr( ~eskI ), !Ltk( $I, ~lkI ) ] --> [ Out( hkI ), Init_1(~eskI, $I, $R, ~lkI ,hkI)]

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rule Init_2:
   [ Init_1( ~eskI, $I, $R, ~lkI ,hkI ), In( Y ) ] --> []
```

Property specification

- `l --[ a ]-> r`
- Actions stored as (action) trace

  Additionally: adversary knows facts K()

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rule Init_2:
  let exI = h1(< ~eskI, ~lkI >),
      key  = h2(< Y^~lkI, pkR^exI, Y^exI, $I, $R >)
  in
   [ Init_1( ~eskI, $I, $R, ~lkI , hkI), In( Y ), !Pk($R,pkR) ]
   --[ ]-->
   []
```

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   --[ Accept(~eskI, $I, $R, key) ]-->
   []
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   --[ Accept(~eskI, $I, $R, key) ]-->
   []


Lemma trivial_key_secrecy:
  "(All #i Test A B key. Accept(Test,A,B,key)@i => Not (Ex #j. K(key)@j ))"
```

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```
rule Ltk_reveal:
   [ !Ltk($A, lkA) ] --[ LtkRev($A) ]-> [ Out(lkA) ]
```

Property specification

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```
rule Ltk_reveal:
   [ !Ltk($A, lkA) ] --[ LtkRev($A) ]-> [ Out(lkA) ]


lemma key_secrecy:
  /* If A and B are honest, the adversary doesn't learn the session key
   */
  "(All #i1 Test A B key.
    (
      Accept(Test, A, B, key) @ i1
      &
      not ( (Ex #ia . LtkRev( A ) @ ia )
          | (Ex #ib . LtkRev( B ) @ ib )
          )
    )
    ==> not (Ex #i2. K( key ) @ i2 )
  )"
```

eCK security model for key exchange

- Adversary can
  - learn **long-term keys**,
  - learn the **randomness** generated in sessions,
  - learn **session keys**
- But only as long as the Test session is ***clean***:
  - **No reveal of session key of** Test session or its **matching session**, and
  - No reveal of randomness of Test session as well as the long-term key of the actor, and
  - If there exists a matching session, then something is disallowed
  - If there is no matching session, then something else...

Specifying eCK

```
Lemma eCK_key_secrecy:
  "(All #i1 #i2 Test A B key. Accept(Test, A, B, key) @ i1
                              & K( key ) @ i2 ==>
  (
  )"
```

Specifying eCK

```
Lemma eCK_key_secrecy:
  "(All #i1 #i2 Test A B key. Accept(Test, A, B, key) @ i1
                              & K( key ) @ i2 ==>
  (
      (Ex #i3. SesskRev( Test ) @ i3 )
    |
  )"
```

Specifying eCK

```
Lemma eCK_key_secrecy:
  "(All #i1 #i2 Test A B key. Accept(Test, A, B, key) @ i1
                                & K( key ) @ i2 ==>
  (
      (Ex #i3. SesskRev( Test ) @ i3 )
    | (Ex MatchingSession #i3 #i4 ms.
          ( Sid ( MatchingSession, ms ) @ i3
          & Match( Test, ms ) @ i4)
          & (Ex #i5. SesskRev( MatchingSession ) @ i5 ))

  )"
```

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    | [ ...andsoforth... ]
  )"
```
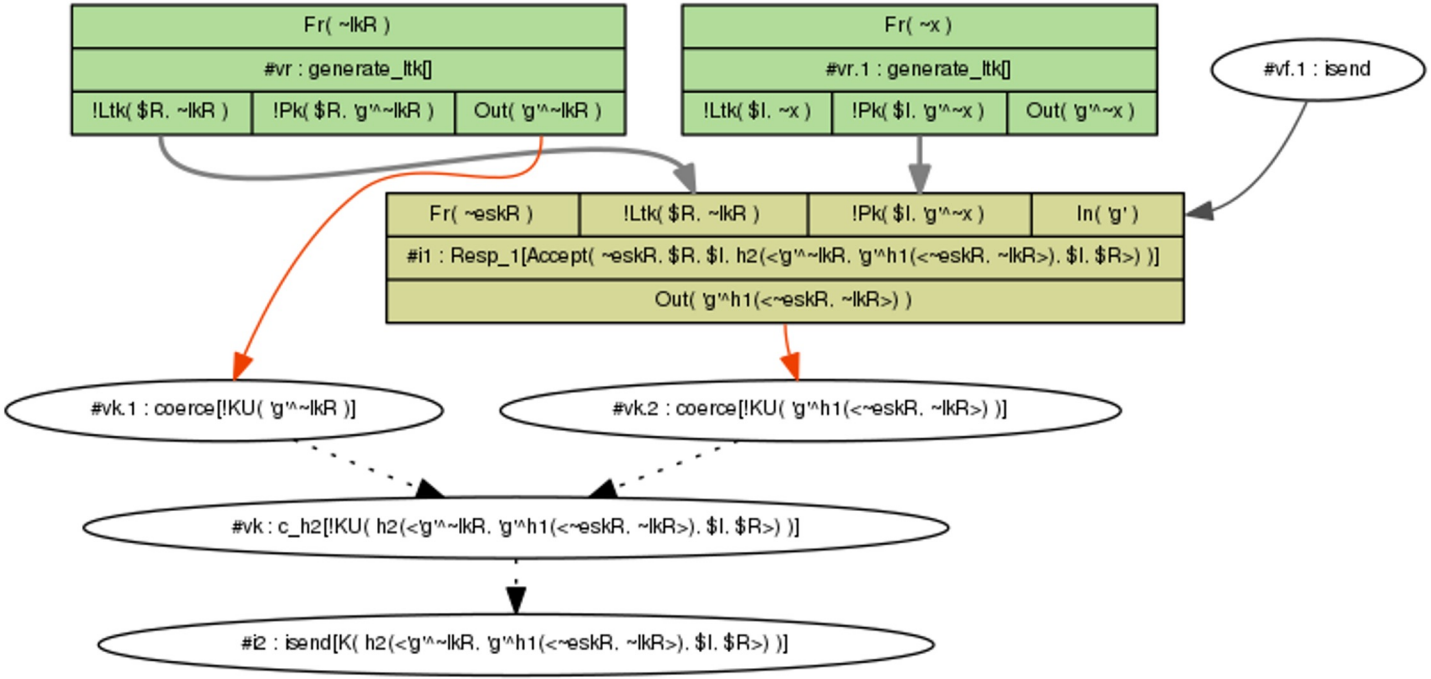

If Test accepts and the adversary knows key, then the Test must not be fresh, i.e., "... **reveal of session key of Test session** or **its matching session**", or ...

Demo

Reading Tamarin's graphs



Tamarin’s algorithm

Basic principles

- Backwards search using **constraint reduction rules** (>25!)
- Turn negation of formula into set of constraints
- Case distinctions
  - E.g.: Possible sources of a message or fact
- Each step of the Proof, solve goals:
  - Solve Premises based on Dependency Graphs
  - Deconstruction Chains
  - Solve Action goals, …

Lemmas

- When it doesn’t terminate…
- Guide the proof manually; export
- Write **lemmas**
  - “**Hints**” for the prover
    - They don't change the guarantees, only help tool in finding a proof
  - E.g. specify lemma that can be used to prune proof trees at multiple points

How do I know my model is correct?

- **It is easy to model something incorrectly**
- Executability: try to prove expected traces actually exist
- Break the protocol on purpose
- Much easier to check these things than in manual proofs!

Tamarin: Conclusions

- **Tamarin** offers **many unique features**
  - Unbounded analysis, flexible properties, equational theories, global state
  - Enables automated analysis in areas previously unexplored
- It has many **other features** I didn't touch on now
  - Induction, restrictions, reusable lemmas, heuristics, ...
  - Many new features planned!
- Tool and sources are **free**; development on Github

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What's next

- Please install Tamarin if you have not done so yet.
- Look at the NAXOS files or the exercises:
  - https://github.com/jdreier/tamarin-90min
  - Ask us for help!

Heuristics?

- If Tamarin terminates, one of two options:
  - **Proof**, or
  - **counterexample** (in this context: attack)
- At each stage in proof, multiple constraint solving rules might be applicable
  - Similar to “how shall I try to prove this?”
  - Choice influences speed & termination, but not the outcome after termination
- Complex **heuristics choose rule**
  - user can give hints or override

Symbolic vs Computational?

Modeling real-world objects

Reality

Symbolic

Modeling real-world objects

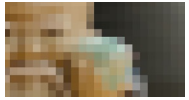
Reality

Computational

Symbolic

Modeling real-world objects

Reality



Computational

Symbolic

References

- Tamarin Website (https://tamarin-prover.com/)
  - Notably links to: all sources, example files, mailing list/google group, manual, tutorial data, (incomplete) list of papers
- More accurate modeling of cryptography
  - *Seems Legit: Automated Analysis of Subtle Attacks on Protocols that Use Signatures*
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  - *The EMV Standard: Break, Fix, Verify*
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